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Cretaceous palaeomagnetic results from Murge (Apulia, southern Italy): tectonic implications

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SUMMARY

The Adriatic area is of key importance in the reconstruction of the kinematic history of the Mediterranean region. Concerning its relative position with respect to the African plate, palaeomagnetic studies (Márton & Veljovic 1983; VandenBerg 1983) have found evidence that 'autochthonous' Istria and Gargano rotated counter-clockwise with respect to Africa, after the Cretaceous. However, the geodynamic role of the southern part of Apulia has remained unclear. In order to decide whether the boundary between the Adriatic microplate and Africa may be drawn between Gargano and Murge and whether the different parts of Apulia could have undergone differential rotations after the Cretaceous, we carried out palaeomagnetic investigations in Murge and the Salentina Peninsula.

In the field, several carefully selected facies were collected and oriented with a magnetic compass at eight geographically distributed localities (46 sites, 224 cores and 13 hand samples).

Because of extremely weak NRM, several sites failed to yield a meaningful palaeomagnetic signal. This category comprises all sites younger than the Turonian bauxite horizon and also six sites of Cenomanian–early Turonian age. Samples from the remaining sites were subjected to detailed thermal and AF demagnetization, Kirschvink (1980) and subtracted vector analysis, and IRM experiments, including the thermal demagnetization of three-component IRM.

As a result, we were able to demonstrate that some of the sites are devoid of overprint. On the other hand, there were sites with unremovable overprint. The latter were rejected from tectonic evaluation, despite the sometimes high degree of clustering of the individual NRMs within a site; so were those with unstable remanence.

Eventually, a Cenomanian–Turonian palaeomagnetic mean direction was calculated for Murge on a locality (before tectonic correction $D = 327°$, $I = 44°$, $k = 55$, $\alpha_{95} = 17°$, $N = 3$, after tectonic correction $D_C = 327°$, $I_C = 38°$, $k = 76$, $\alpha_{95} = 15°$) and a site (before tectonic correction $D = 324°$, $I = 41°$, $k = 44$, $\alpha_{95} = 6.5°$, after tectonic correction $D_C = 327°$, $I_C = 40°$, $k = 42$, $\alpha_{95} = 6.7°$) basis and a palaeomagnetic pole (latitude 56°, longitude 261°) was computed from the tectonically corrected locality mean direction.

Since coeval palaeomagnetic poles for Murge, Gargano and Istria cluster, we conclude that the stable part of the Adriatic region moved in a highly integrated manner. Their displacement with respect to Africa has manifested itself in a moderate CCW rotation since the Turonian.

Key words: Apulia, Cretaceous, palaeomagnetism.

INTRODUCTION

The Adriatic area is very important for the reconstruction of the kinematic history of the Mediterranean region. The papers dealing with the geodynamic role of the Adriatic area see it either as an African promontory, still attached rigidly to Africa (e.g. Argand 1924; McKenzie 1972; Channell, D'Argenio & Horváth 1979; D'Argenio & Horváth 1984; Dewey *et al.* 1989) or as a microplate (Lort 1971; VandenBerg & Zijderveld 1982; Morelli 1984; Anderson &

Jackson 1987; Jongsoma *et al.* 1987; Márton 1987; Ricchetti *et al.* 1988; Favali, Mele & Mattietti 1990; Westaway 1990). As well as this basic incompatibility of interpretations, there is no general agreement about the size of the Adriatic microplate (Lort 1971; Morelli 1984; Anderson & Jackson 1987; Jongsoma *et al.* 1987; Ricchetti *et al.* 1988; Westaway 1990).

Palaeomagnetic data relevant to this problem have been accumulating since the late 1970s. First, Channell & Tarling (1975) reported Cretaceous data from Apulia suggesting that the Adriatic was autochthonous and is still rigidly connected to the African plate. However, the palaeomagnetic results that tied Adria to Africa were far from unambiguous, since they were obtained from bauxites (Channell & Tarling 1975) which are extremely liable to acquire remanence during secondary processes. Later, Channell (1977) published results that seemed to support those obtained from the bauxites. Unfortunately, the rocks sampled by him were too weakly magnetic for adequate cleaning in the laboratory.

Soon after, the relative movement of Adria with respect to Africa was demonstrated for two autochthonous parts of the Adriatic region (Fig. 1)—Istria (Márton & Veljovic 1983) and Gargano (VandenBerg 1983)—in the way of moderate counter-clockwise rotation.

The Cretaceous of the Murge and Salento area (Fig. 1) has also been tested by Tozzi (1989), but no results were obtained. On the contrary, well-defined magnetization directions were reported for sediments of Palaeogene age from several points in the Gargano and Salento areas (Tozzi *et al.* 1988; Tozzi 1989). These authors stated that Gargano and Salento rotated in post-Oligocene times, both in the clockwise sense with respect to Africa, but by different angles.

The late and early Cretaceous palaeomagnetic poles, respectively, for Istria and Gargano are nearly coincident.

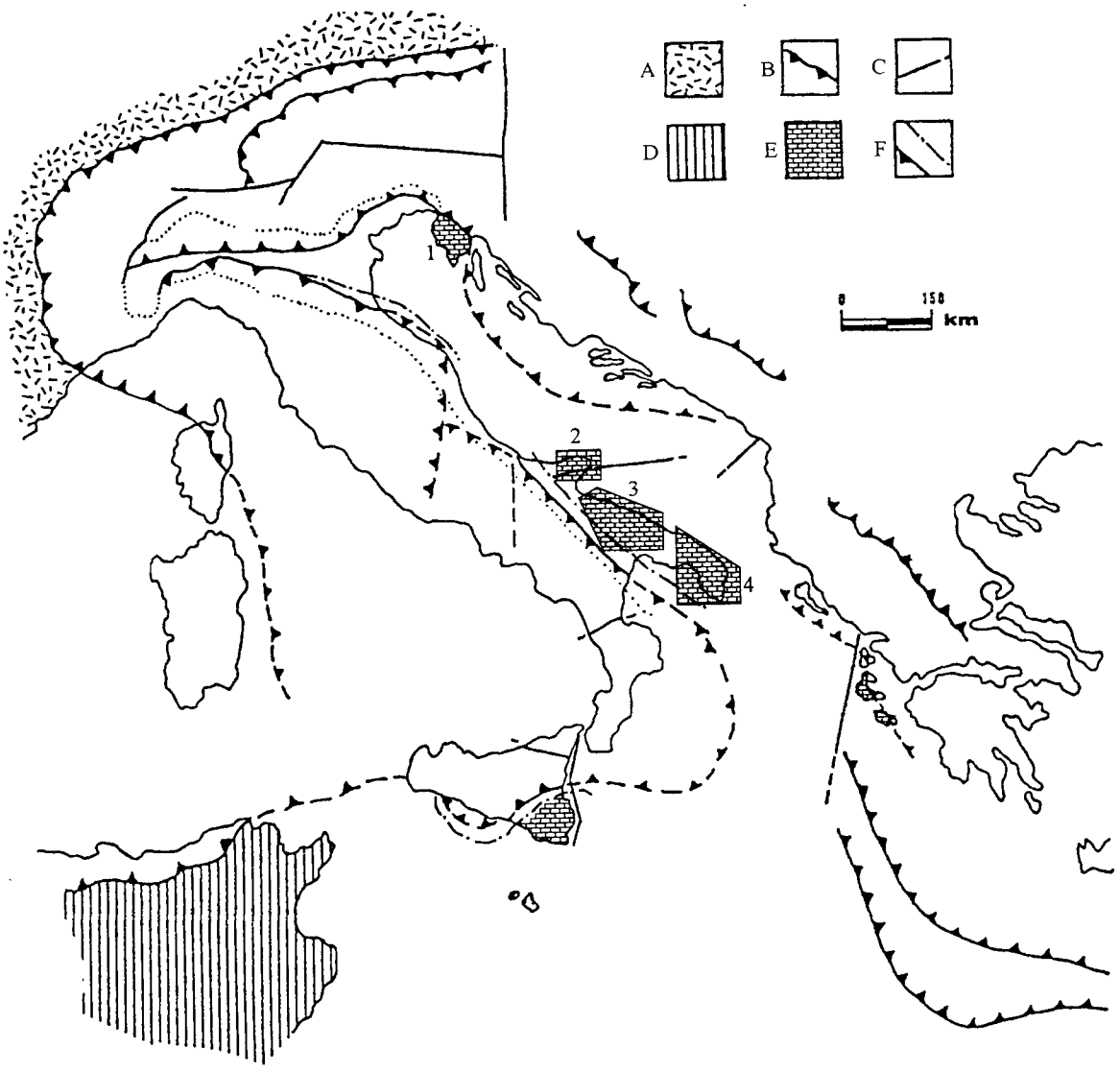


Figure 1. Main tectonic features of the Adriatic area with the autochthonous Istria (1), Gargano (2), Murge (3) and Salento (4). Key: European foreland (A), overthrust front (B), main faults (C), African foreland (D), Adriatic foreland (E), Padano–Bradanic foredeep (F).

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Thus, we have no reason to assume relative motion between them in the Cenozoic. However, the same may not be true for the southern part of Apulia, since Murge and Salento are situated south of important tectonic boundaries (Fig. 1). These are even interpreted by some (e.g. Anderson & Jackson 1987; Favali *et al.* 1990; Westaway 1990) as the contact between the Adriatic microplate and Africa.

The aim of this study was to define Cretaceous palaeomagnetic poles for Murge and Salento Peninsula, thus seeking answers to the following questions: (1) have the different parts of Adria really undergone differentical rotations in the Cenozoic? (2) Is it possible to draw the boundary between the Adriatic microplate and Africa between Gargano and Murge? Since we believe that in times of fast apparent polar wander, the correct answer to the problem of relative movement may be found by comparing palaeomagnetic poles of precisely the same age (Márton 1993), we collected rocks matching in age those already yielding positive results in Istria and Gargano.

GENERAL GEOLOGY AND SAMPLING

Murge and Salento (Figs 1 and 2) belong to the Mesozoic–Tertiary Apulian Domain extending from Gargano Promontory to the Kefalonia–Zante escarpment. It is bordered by the late Tertiary Bradanic basin in the west and the Otranto–Lefkas basin in the east. This domain, a Mesozoic carbonate platform, was the foreland area to both the Apennines and Dinarides during the Neogene (Ogniben 1969; D'Argenio, Prescatore & Scandone 1973).

In the Murge and the Salento Peninsula the majority of the outcrops are of Cretaceous age. The Cretaceous shallow-water carbonates are divided by a late Turonian bauxite horizon into an older Calcare di Bari Formation and a younger Calcare di Altamura Formation (Ricchetti 1975; Iannone, Sinni & Pieri 1979; Iannone & Laviano 1980). Dolomitization occurs in places.

The Capitanata Graben forms the northern boundary of the Murge, separating it from Gargano (Fig. 2). North and south of this graben the thickness of the crust is markedly different: in Gargano the Moho is observed at 25–30 km, while in the Murge it is at 35–40 km (Morelli & Nicholich 1988). In the south, the Brindisi Graben separates Murge from the Salentina Peninsula (Fig. 2). This graben, however, is a less significant tectonic feature than the Capitanata, since the structure and thickness of the crust remains unchanged on either side of it.

The platform sequence is gently folded into dominant NW–SE striking structures, plunging towards SE (Ricchetti *et al.* 1988). Early Cretaceous rocks are found outcropping in the north-west only; Senonian and Tertiary rocks at the southern corner. The deformation predates the deposition of the bauxites (Campobasso, Richetti & Luperto Sinni 1972).

We sampled several carefully selected shallow-water limestone facies, representing the Cenomanian–Senonian age interval, i.e. both the Calcare di Bari and the Calcare di Altamura formations were tested. The total number of 224 cores drilled and oriented in the field and the 13 block samples represented seven localities in Murge (six of them devoid of dolomitization) and one locality in the Salentina Peninsula (with evidence for dolomitization followed by dedolomitization). From the block samples, oriented in the field, cores were drilled in the laboratory.

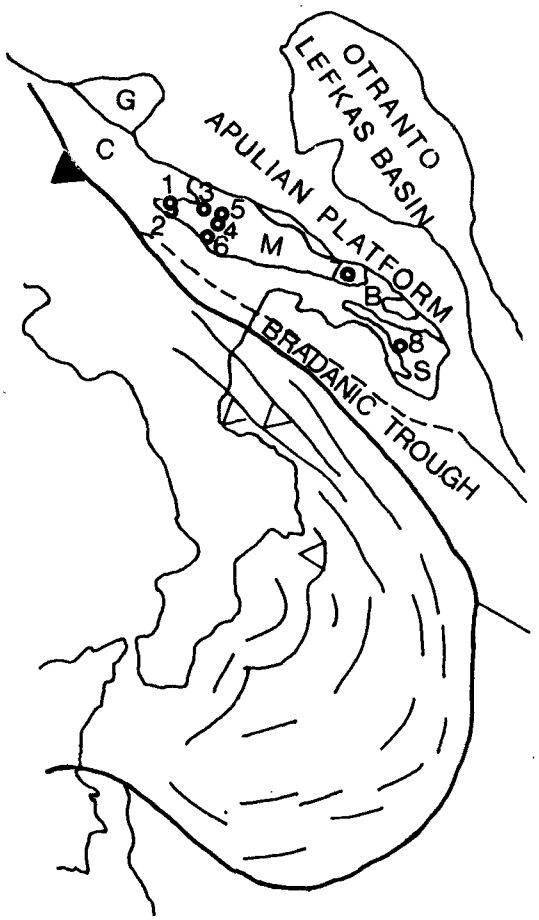


Figure 2. Sampling area with eight Cretaceous test localities. C, Capitanata Graben; B, Taranto–Brindisi Graben. Test localities: Monte Scorzone, early–mid Cenomanian (1); Spinazzola, Cenomanian–Turonian (2); Martine di Iatta, Cenomanian–Turonian boundary (early Senonian?) (3); Bivio, Cenomanian–Turonian boundary (4); Pezza Rossa, Cenomanian–Turonian boundary (5); Altamura, Senonian (6); Messapico, Senonian (7); Veglie, Senonian (8).

The sampled localities were quarries or road cuts that do differ both in facies and age. Monte Scorzone is the oldest (early–mid Cenomanian) and represents cyclic sequences of back-reef limestones. The massive limestones of Spinazzola and the near-reef facies of Martina di Iatta and Bivio are slightly younger. Pezza Rossa is situated at the top of the Calcare di Bari Formation. The localities sampled in the Calcare di Altamura Formation are situated far apart. They are more uniform lithologically than the collected members of the Calcare di Bari Formation. Nevertheless, they differ in age quite significantly.

At each locality we sampled several beds, which are regarded as different sites.

LABORATORY PROCESSING AND RESULTS

The samples were studied in the palaeomagnetic laboratory of the Eötvös Loránd Geophysical Institute in Budapest.

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The remanance was measured with GM400 cryogenic and JR-4 spinner magnetometers. Sister specimens of selected samples were demagnetized in increments, one specimen in AF field, up to a maximum of 0.24 T, the other in a Schonstedt oven. Susceptibility change on heating was monitored using a KLY-2 bridge.

On AF demagnetization we observed instability (e.g. Fig. 3, AP42A) or extremely hard behaviour (e.g. Fig. 4, AP2B). Complete demagnetization was achieved by thermal treatment during which susceptibility was fairly stable till the NRM signal was lost. Therefore, all the non-pilot samples were thermally demagnetized at 150 °C, 200 °C, then in 25 or 50 °C increments. The characteristic magnetization of each sample was defined from several consecutive clening steps by applying Kirschvink (1980) analysis.

Some specimens, earlier treated in AF field, were given

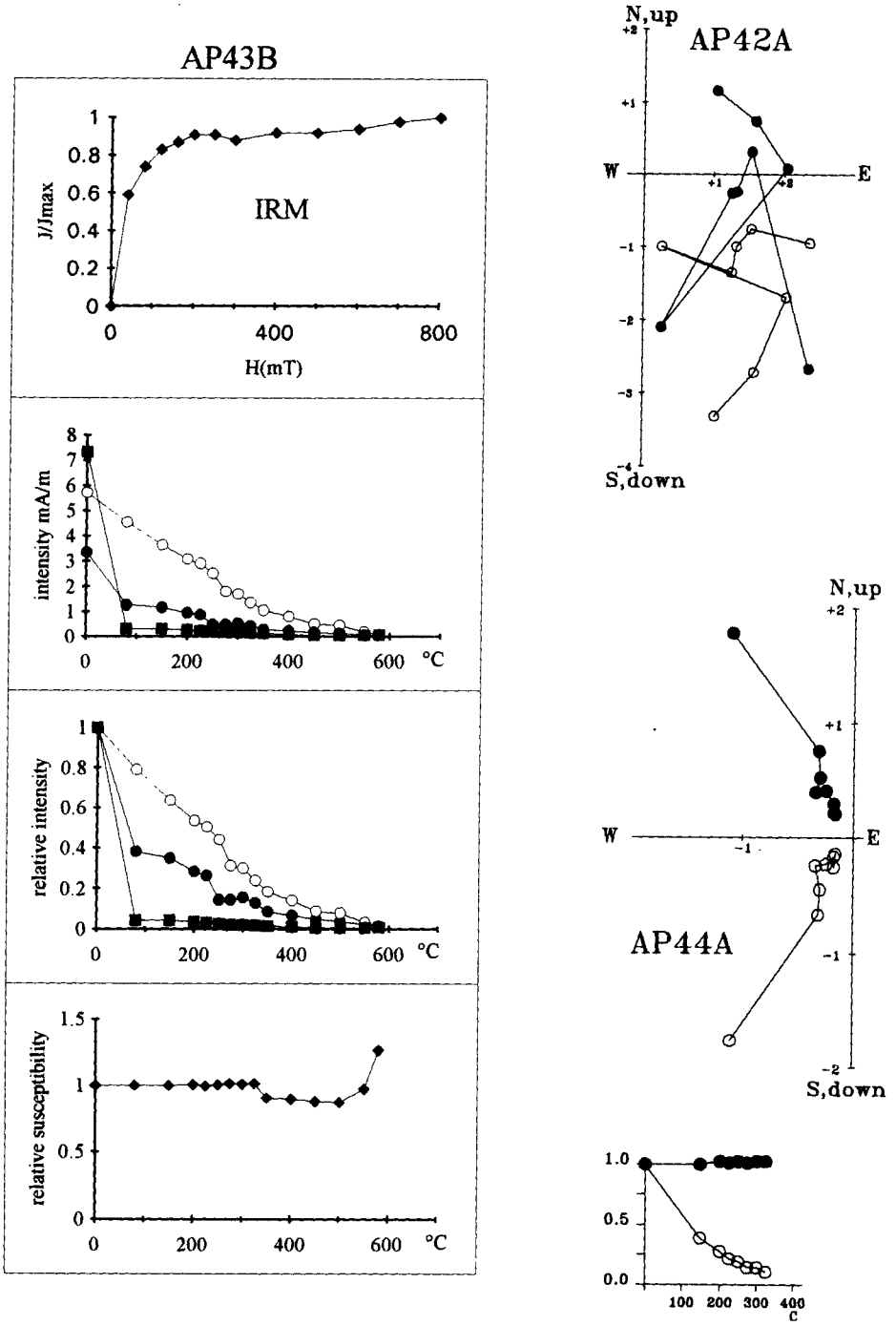


Figure 3. Monte Scorzone. Left side, from top to bottom: typical IRM acquisition curve (maximum intensity 6.6 mA m^{-1}); unblocking behaviour of the soft (open circles), medium (solid circles) and hard (squares) components of the IRM; and normalized susceptibility versus temperature curve. Right side, from top to bottom: characteristic behaviour of NRM on AF (AP42A: 1 unit = 4×10^{-5} A m^{-1} increments of 10 mT) and thermal (AP44A: 1 unit = 4×10^{-5} A m^{-1}) demagnetization (Zijderveld diagrams); normalized intensity (open circles) and susceptibility (solid circles) versus temperature curve (bottom right): initial susceptibility: -3.6×10^{-6} SI.

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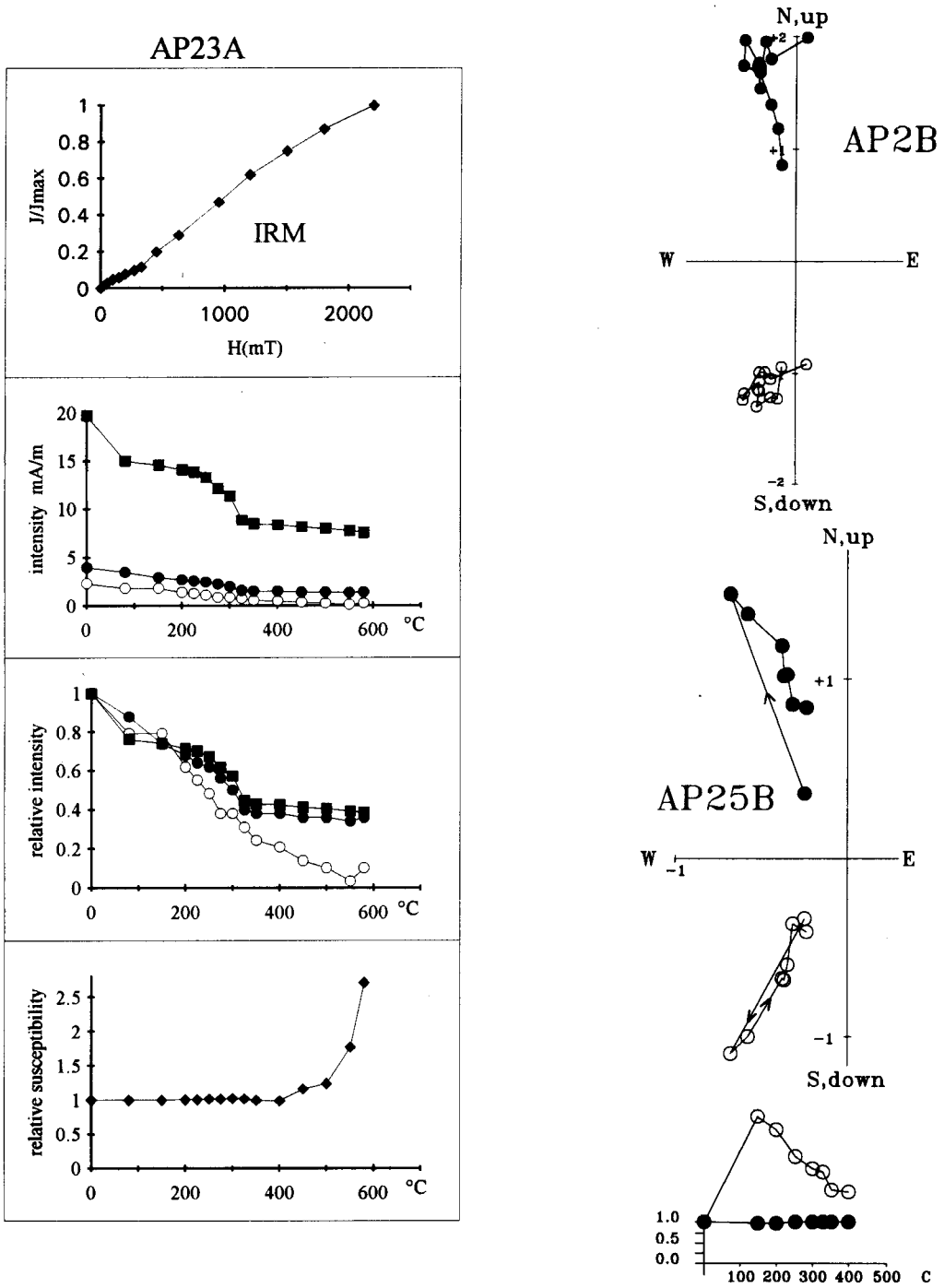


Figure 4. Spinazzola. The diagrams are the same as in Fig. 3 (AP23A: maximum IRM intensity: 31 mA m^{-1} AP2B and AP25B: units are 4×10^{-4} A m^{-1} and 1.6×10^{-5} A m^{-1}, respectively; AP25B: initial susceptibility 4×10^{-6} SI). AP2B demagnetization steps: increments of 10 mT up to 60 mT followed by 80, 100, 130, 160, 200, 135 mT.

IRM with a pulse magnetizer (1.0 T peak field) followed by magnetization with an electromagnet (peak field 2.3 T).

In addition to IRM acquisition characteristics, the behaviour of the hard (saturating between 0.4 and 2.3 T), medium (saturating between 0.12 and 0.4 T) and soft (saturating below 0.12 T) components was studied during thermal demagnetization (Lowrie 1990).

Comparison between the behaviour of the NRM and the susceptibility on heating on the one hand, and IRM acquisition characteristics and the unblocking properties of the three components of the IRM on the other hand, served as a basis for identification of the actual and potential carriers of the remanence. (Figs 3–8).

In all studied specimens the NRM signal was lost below the Curie point of the magnetite, while the susceptibility remained basically unchanged below 400 °C.

Based on the behaviour of the susceptibility on heating, the presence of maghemite in our samples can be safely

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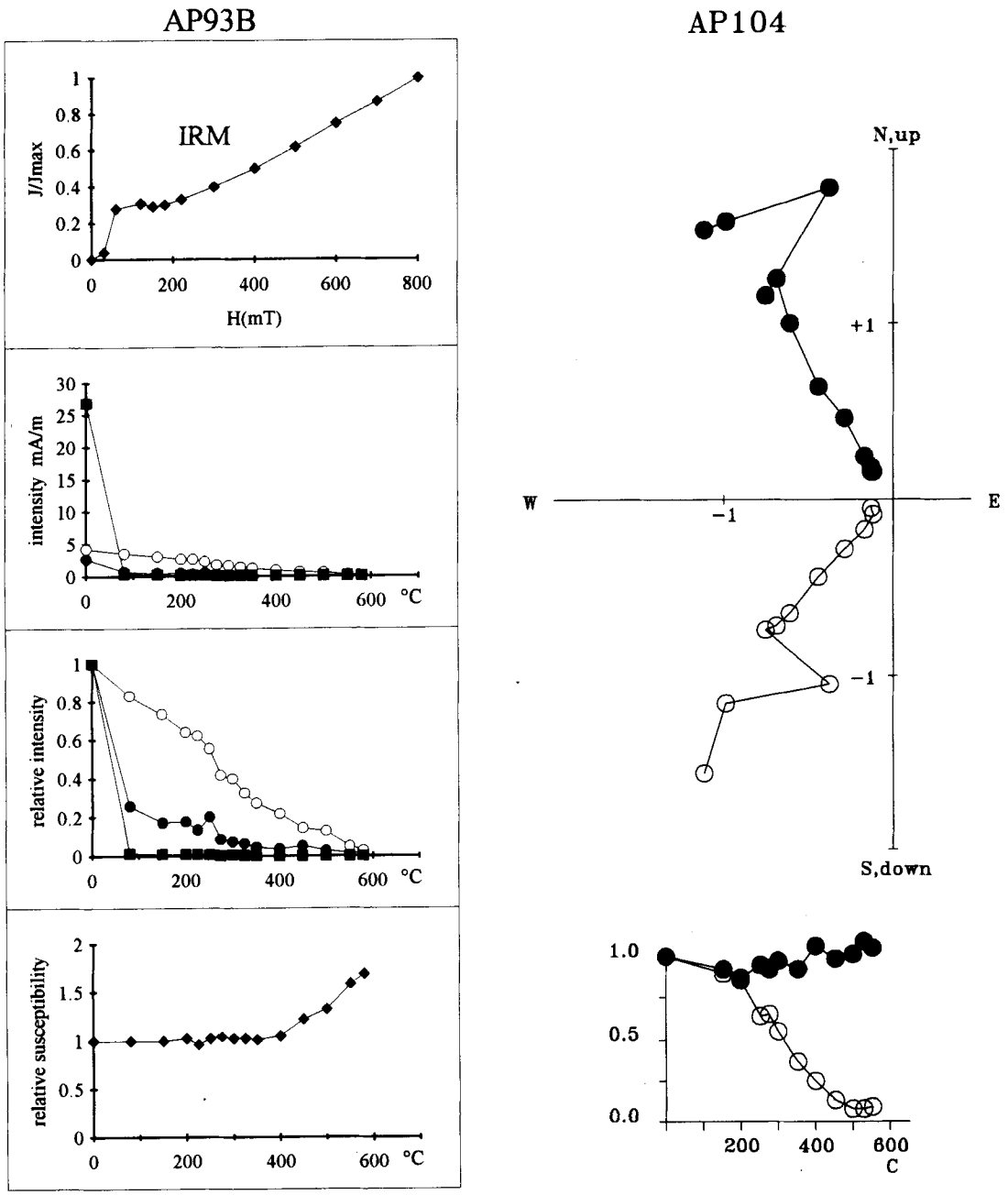


Figure 5. Martine di Iatta. Left side: same as in Fig. 3 (AP93 maximum IRM intensity: 153 mA m^{-1}). Right side: characteristic behaviour of the NRM on thermal demagnetization. Top: Zijderveld diagram (1 unit = 4×10^{-5} A m^{-1}), bottom: normalized intensity (open circles) and susceptibility (solid circles) as a function of temperature (initial susceptibility: -11×10^{-6} SI).

ruled out. We came to a similar conclusion concerning sulfides. The reasons are as follows.

(1) Though the unblocking spectrum of both the NRM and the soft component of the orthogonal IRM (and often that of the medium component), show prominent unblocking features well below the temperature of the final decay, the kink(s) of the unblocking spectra normally appear at 200–300 °C instead of 300–400 °C. (In order to distinguish between unblocking and fluctuation due to the extremely small value of the IRM component, both the intensities and the normalized intensities are shown in Figs 3–8 as a function of temperature.)

(2) Susceptibility is stable between 200 and 400 °C, while iron sulfides carrying remanence definitely show susceptibility variation in the same temperature interval (see e.g. Dekkers 1990).

(3) Susceptibility does increase from 400–500 °C on, but a decrease is not observed above 550 °C as it should if the carrier were, for example, pyrrhotite (Dekkers 1990).

The easily saturating phase of the IRM acquisition curves, therefore, is thought to be magnetite. The low unblocking property of the magnetite phase of both the NRM and the IRM may be explained by small grain size (Aïssaoui, McNeill & Kirschvink 1990).

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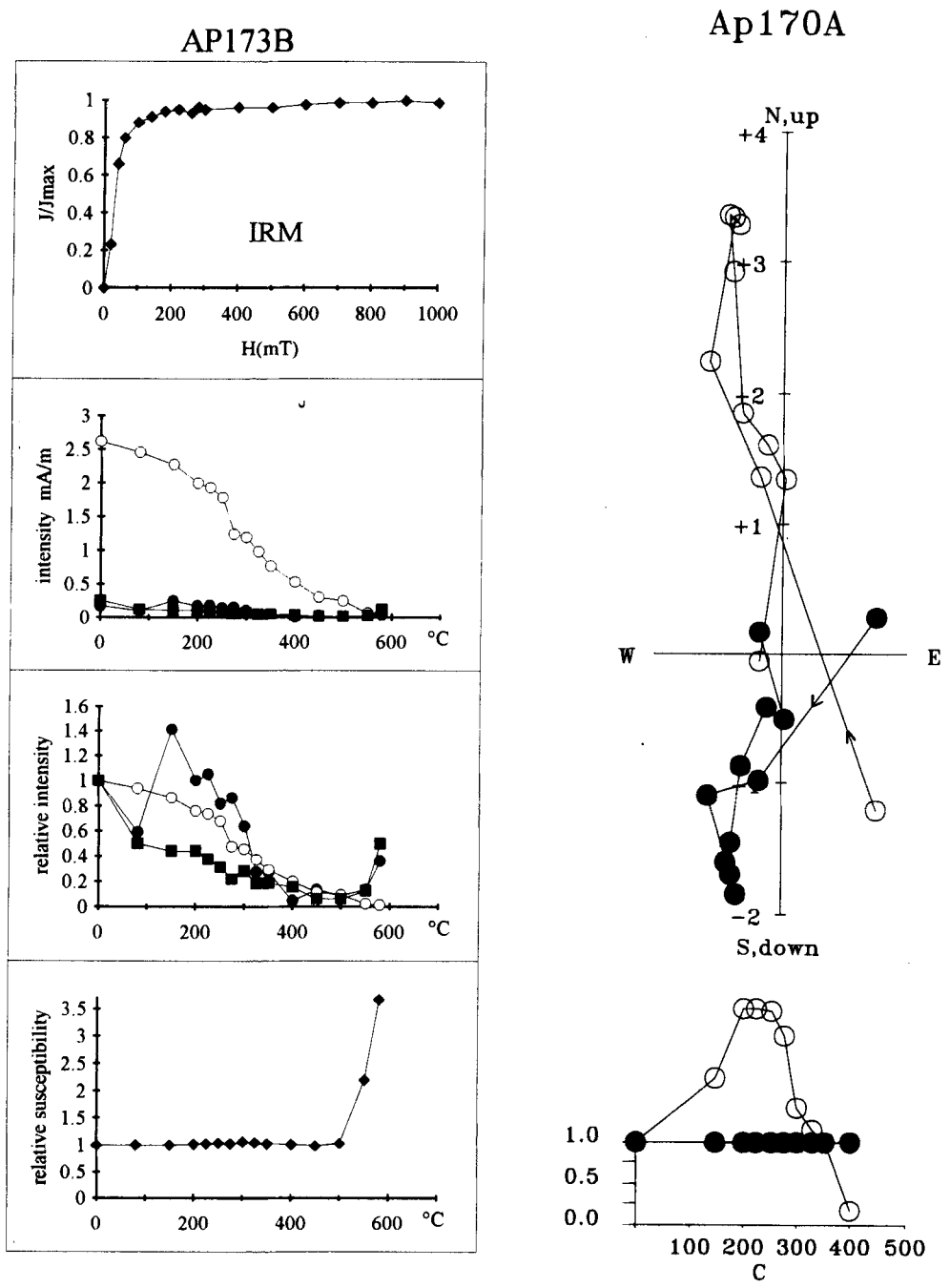


Figure 6. Bivio. Left side: same as in Fig. 3 (AP173B: maximum IRM intensity: 2.7 mA m^{-1}); right side same as in Fig. 5 (top: 1 unit = 8×10^{-6} A m^{-1}; bottom: initial susceptibility: 10.5×10^{-6} SI).

Most of the IRM acquisition curves indicate the presence of medium and hard magnetic phases. In some cases (Figs 5 and 8: sample AP192B) these proved to be basically goethite; in others (Figs 4 and 8: sample AP195B) dominant haematite (a phase surviving the Curie point of the magnetite) is indicated by the unblocking spectra of the medium and hard components of the orthogonal IRM.

Since the decay of the NRM in all investigated samples is complete before 575 °C, in estimating the contribution of haematite to the NRM we are guided by the IRM acquisition character, plus the proportion and unblocking behaviour of the orthogonal components of the IRM.

On a general level, we found that the platform carbonates of our collection are basically of two types: one group contains magnetite plus goethite (Figs 3 and 8: sample AP192B); the other magnetite plus haematite and little goethite (Figs 4 and 8: sample AP195B). Locality 3 (Fig. 2) is the only one where magnetite seems to be the only magnetic mineral present.

Overprint from goethite is easily removed by heating; therefore we payed special attention to the problem of the contribution of haematite to the NRM, which we believed secondary for two reasons. The first is that the low unblocking temperature is a contra-indication of detrital

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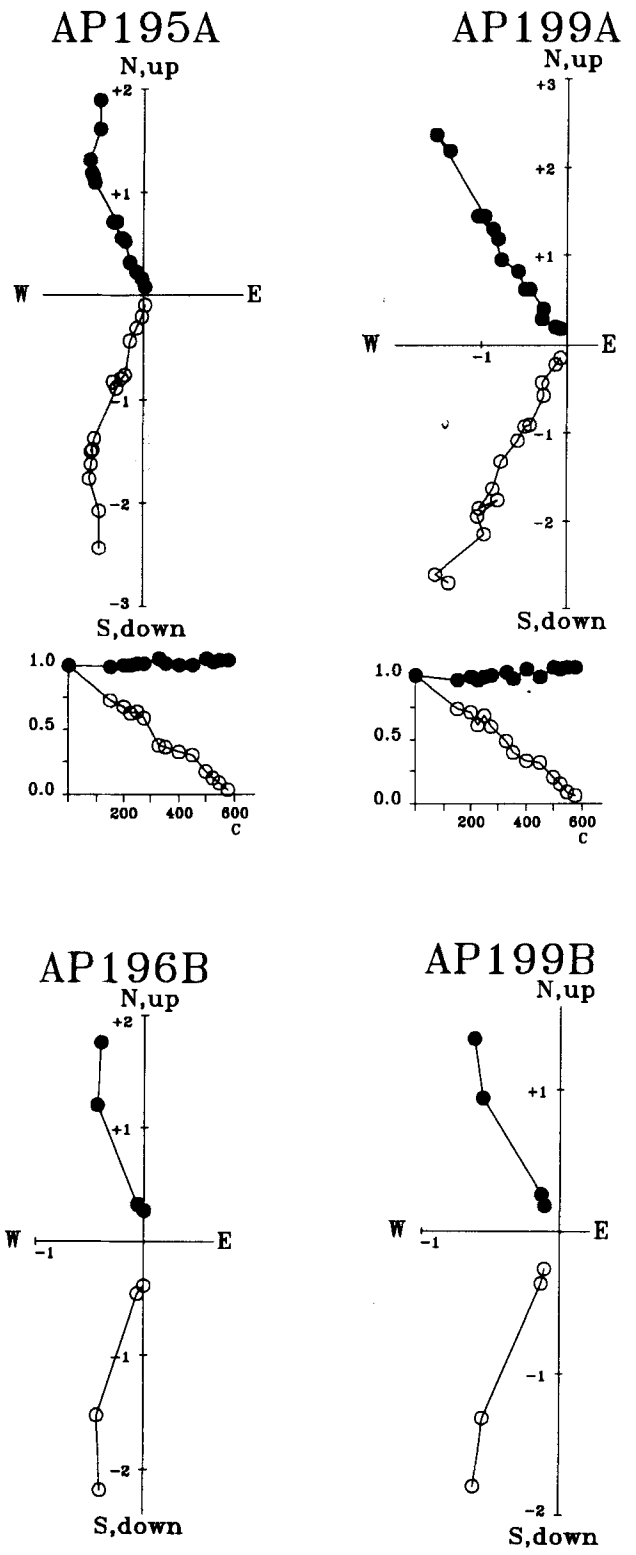


Figure 7. Pezza Rossa. From top to bottom: the behaviour of the NRM on thermal demagnetization, Zijderveld diagrams (units are 1.6×10^{-4} A m^{-1} and 4.0×10^{-5} A m^{-1}, respectively); normalized intensity and susceptibility versus temperature curves (initial susceptibilities are -9.5×10^{-6} SI and -11×10^{-6} SI, respectively) and the behaviour of the NRM on combined AF and thermal demagnetization (steps are NRM, 150 °C, 40 and 60 nT, units are 1.6×10^{-4} A m^{-1} and 4×10^{-5} A m^{-1}, respectively). Left side: red; right side: with sample.

speculrite; the other is that haematite seems to be a secondary dehydration product of the goethite.

Calcare di Bari Formation

Most sites from this formation yielded a statistically well-defined palaeomagnetic direction. These sites were further scrutinized for possible overprinting. The results are summarized for each locality as follows.

Monte Scorzone

During AF cleaning the NRM was extremely unstable. On thermal demagnetization the intensity reached the noise level of the magnetometers at about 350 °C. In the 150–325 °C temperature range (Fig. 3) characteristic magnetization was isolated for six (Table 1) of the collected 11 sites, distributed over an area of more than a square kilometre and representing 100 m stratigraphically.

IRM acquisition was very fast below 0.2–0.3 T. Above this field the intensity of the IRM did not increase up to 0.8 T (Fig. 3). In higher fields, harder components started to acquire remanence. However, the instability of the soft component in strong magnetic fields resulted in a high fluctuation of the IRM values, superimposed on an increasing saturation remanence.

In addition to the experiments documented in Figs 3–8, we attempted Curie-point measurement on one specimen (Fig. 9). The first half-cycle, despite the high noise, clearly shows that if susceptibility changes on heating, there is a decrease around the Curie point of the magnetite (Fig. 9) and susceptibility increase, indicating newly formed magnetite, starts at much higher temperatures as it would if iron sulfides had decomposed.

Based on the above observations and the arguments in the previous section, Monte Scorzone belongs to the magnetite plus geothite type of platform carbonate, with easily removable overprint.

Spinazzola

NRM was extremely stable on AF demagnetization. During thermal demagnetization different sites behaved differently: at some, goethite appeared to be an important carrier of the NRM; at others it was insignificant (Fig. 4).

IRM acquisition curves exhibited a small soft but strong medium and hard components. Of the three, the first completely decayed by 500 °C, while the others survived 580 °C (Fig. 4). These characteristics classify Spinazzola as a magnetite plus haematite (and goethite) type platform carbonate.

During demagnetization, the declination of the NRM was close to zero in some samples; in others, it had moved first to the west, then returned towards zero (Fig. 4).

In the light of the rock magnetic observations, the directional change may be explained by the overlap of the unblocking spectra of an ancient and a secondary component (the latter along the present Earth's field). Separation of the two is impossible, for the overprint component seems to be present in all coercivity and unblocking range and removed vectors are too weak to be useful.

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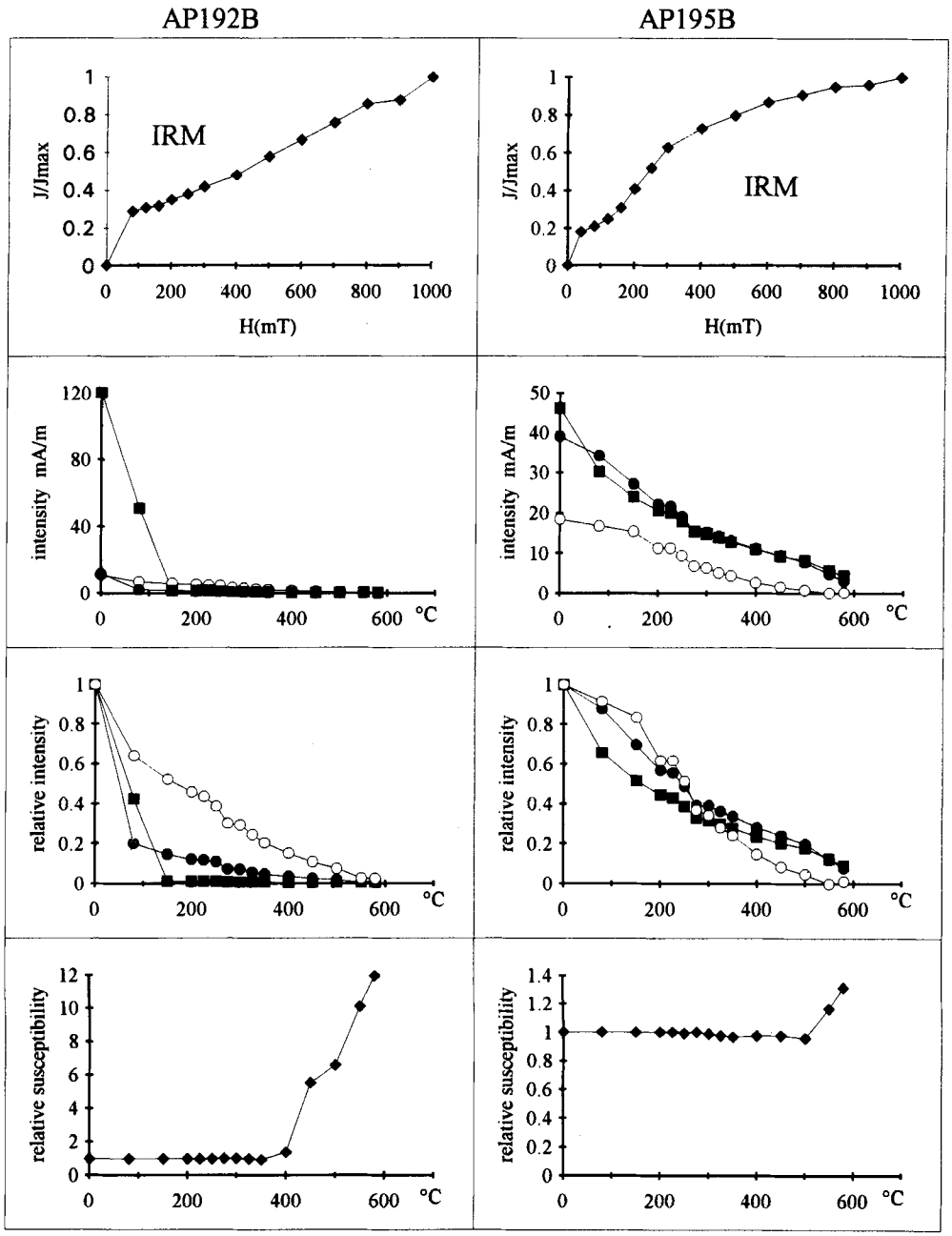


Figure 8. Pezza Rossa. IRM characteristics of a white (left) and a red (right) specimen. Arrangement of the diagrams corresponds to the left side of Fig. 3 (maximum IRM intensities are 39 and 73 mA m^{-1} respectively).

Martine di Iatta

From the huge quarry at Martine di Iatta, six sites were collected at different points and markedly different stratigraphic levels. The last two of them (Table 1, sites 107–111 and 118–123) are suspected to follow the first ones after a considerable hiatus which could correspond to the late Cretaceous bauxite horizon, separating the Calcare di Bari and Calcare di Altamura formations. Nevertheless, the samples (except the first site, which was a failure) behaved in a highly consistent manner during laboratory investigations, justifying a common characterization of all sites.

NRM intensity decreased continuously on heating, while a slight decrease, followed by moderate increase above 400 °C, was characteristic of the susceptibility. An apparently single component was isolated from 250 °C onward. This part of the NRM is basically connected to magnetite, since the IRM unblocking diagrams indicate that no haematite is present (Fig. 5).

This locality, therefore, belongs to the magnetite plus goethite type.

Bivio

Seven sites were collected from different beds of a road cutting: six from well-bedded white limestone and one from

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Table 1. Murge: summary of palaeomagnetic directions.

Sampling locality	N	D° Dc°	I° Ic°	k	α95°
MONTE SCORZONE AP 34-78, 124-147					
Site 42-46	5	320	33	23	16
		318	28		
Site 47-50	4	324	51	166	7
		320	46		
Site 55-58	2	320	43	-	-
		317	38		
Site 59-65	4	327	38	41	15
		327	38		
Site 130-134	5	311	50	73	9
		317	54		
Site 135-140	6	301	51	19	16
		306	56		
Locality mean	6	318	45	65	8
		318	44	48	10
MARTINE DI IATTA AP 79-123					
Site 88-94	6	334	36	340	4
		339	39		
Site 95-100	6	338	43	250	4
		344	45		
Site 101-106	5	327	30	157	6
		331	34		
Site 107-111	4	333	33	84	10
		337	36		
Site 118-123	3	318	26	57	17
		332	25		
Locality mean	5	330	34	79	9
		336	36	92	8
PEZZA ROSSA AP 192-204					
Site 192-194		332	53	74	8
Site 198-200 combined		326	32	42	10
Sites 34-41, 51-54, 56-78, 79-87, 112-117, 124-129, 141-147 too weak					
Sites 195-197, 201-204 overprinted					

N: Number of good samples; *D*, *I*; *Dc*, *Ic*: palaeomagnetic mean declination (*D*) and inclination (*I*) before and after (*Dc*, *Ic*) tilt correction; *k*, α_{95}: statistical parameters (Fisher 1953).

a breccia layer. One bed was too weakly magnetic, six yielded the results summarized in Table 2. According to the IRM acquisition curve, magnetite is the dominant magnetic mineral at Bivio, for IRM saturates in low field and the intensity does not increase above 0.9 T (compare the maximum IRM intensity on the acquisition curve with the intensity of the IRM components below it). The soft component of the IRM is very strong compared to the medium and hard components (Fig. 6).

This locality is perfect from the viewpoint of magnetic mineralogy, containing practically a single magnetic mineral, magnetite. Nevertheless, there are arguments (see discussion) against the primary origin of the characteristic magnetization at Bivio.

Pezza Rossa (Rosato delle Murge)

Reddish and whitish beds alternate at this locality. The four sampled beds, two of each sort, encompass about 1 m stratigraphically. Reddish samples invariable showed overprinting along the actual field, which was apparently removed at moderate temperatures. Whitish samples were mostly free of overprint (Fig. 7, specimens 195A and 199A,

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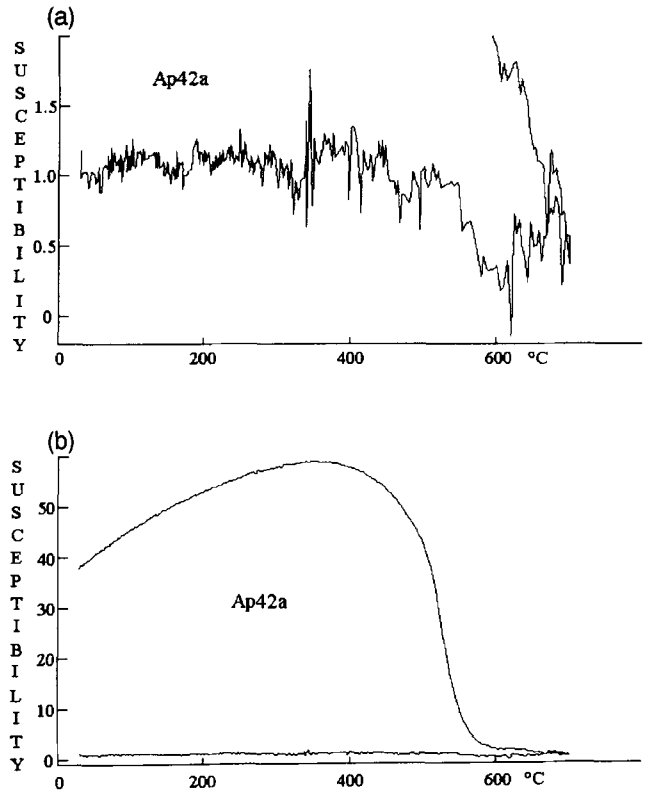


Figure 9. Monte Scorzone (specimen AP42a) Curie-point measurement. Susceptibility is in 10^{-6} SI units. (a) First half-cycle (heating); susceptibility starts to increase on incipient cooling. (b) Heating–cooling cycle showing the formation of magnetite above 600 °C.

respectively). Full demagnetization was achieved by 575 °C, in both cases.

After the apparent removal of the overprint, westerly declinations were measured that were systematically larger in the white than in the red samples (mean direction from red samples: $D = 340°$, $I = 50°$, $k = 273$, $\alpha_{95} = 4°$ before tilt correction, $D = 339°$, $I = 40°$, $k = 103$, $\alpha_{95} = 6$ after tilt correction; for the mean direction for white samples, see Table 1). We suspect that the red samples contain a high blocking overprint component, in addition to the low blocking one.

By changing demagnetization strategy from a dominantly thermal treatment to dominantly AF demagnetization (Fig. 7, lower part) we are able to demonstrate that the NRM of the white samples, after the removal of the overprint from goethite, really became single component. In contrast, the westerly declination of the red limestone must be biased towards the actual field for the declination plot meets the N–S axis above the origin.

The unblocking temperature properties of the three-component IRM of the white (Fig. 8, left side) and red (Fig. 8, right side) samples also revealed that the proportion of the medium and hard components to the soft one is much higher (after 150 °C) in the red than in the white material.

It is interesting to note that the iron minerals of the Rosato della Murge—according to the mineralogical investigations by Maggiore *et al.* (1980)—are haematite and goethite. We think, however, that magnetite must also be present, for the IRM acquisition curves clearly show an easily saturating phase (Fig. 8, top) and a considerable amount of the NRM is lost during AF demagnetization (Fig. 7, bottom).

DISCUSSION

As was mentioned in the previous section, statistically reasonably well-defined palaeomagnetic directions can be identified in samples from the Calcare di Bari Formation.

The characteristic magnetizations at two of these localities, Monte Scorzone and Bivio, might be related to magnetite without the interference of any other magnetic mineral that may survive 150 °C.

The simple remanence at Monte Scorzone has a westerly declination and intermediate inclination, both before and after tilt correction (Table 1). There is some scatter in the inclinations of the site means, which we regard as natural in rocks having very weak remanence. The natural remanence resides in a magnetic phase that exhibits all the characteristics of a fine-grained magnetite, similarly to the carrier mineral in recent carbonate-platform sediments

Table 2. Bivio: summary of palaeomagnetic directions.

		N	D°	I°	k	α95°	Dc°	Ic°	k	α95°
1.site	AP166-68	1	170	-61	-	-	203	-77	-	-
2.site *	169-72	4	182	-58	92	10	215	-71	92	10
3.site	173-75	3	169	-66	85	13	203	-69	85	13
4.site	176-78	3	176	-70	46	18	236	-71	46	18
5.site	179-81	2	184	-64	-	-	223	-65	-	-
6.site **	182-86	3	183	-57	103	12	216	-60	103	12
mean of sites		6	178	-63	193	5	217	-69	128	6

* 1 sample is of normal polarity!
** breccia!

site 1-5 on one side, site 6 on the other side of the road

N: number of good samples; *D*, *I*; *Dc*, *Ic*: palaeomagnetic mean declination (*D*) and inclination (*I*) before and after (*Dc*, *Ic*) tilt correction; *k*, α_{95}: statistical parameters (Fisher 1953).

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(Aïssaoui *et al.* 1990). All these properties suggest that the NRM is primary.

It is interesting to note that the statistical parameters belonging to the locality mean direction are slightly better before tilt correction than after. However, the variation in tilt attitude is very little within the locality. If we take into account the not-too-high statistical precision of the site means, which is normal for weakly magnetic rocks, as well as the fact that determinations of structured tilts are never precise to a degree, one cannot be surprised at the somewhat worsened statistical parameters on tilt correction. Moreover, minor variations in tilt are not necessarily of tectonic origin. Therefore, we do not attribute special importance to moderate improvement or worsening of statistical parameters at within-locality level. What we think to be really important is the behaviour of the statistical parameters at between-locality level.

At Bivio, the palaeomagnetic declination, before tilt correction, is close to the actual field direction (but polarity is dominantly reversed!) and inclination is high even for the present day; on tilt correction, the declination becomes easterly (Table 2).

Although carried by magnetite, we consider the NRM at Bivio secondary for several reasons:

(1) before tilt correction the direction is close to that of the actual field;

(2) after tilt correction, the inclination is too high for the Cretaceous;

(3) the polarity is dominantly reversed, not fitting for rocks magnetized during the Cretaceous quiet interval (Lowrie & Channell 1980);

(4) the breccia layer exhibits the same characteristic remanence as the massive limestone beds.

More complex NRM was found at the other localities. Despite the careful laboratory processing, perfect separation of the ancient remanence from the overprint components was not always possible.

A clear case, with unremovable overprint, is Spinazzola (Fig. 4). Though the mean direction of the NRM was statistically fairly well defined at nearly all cleaning steps, a considerable part of the overprint component along the present field was more stable than the ancient component of the NRM. For each sample, however, there was a cleaning step where the declination reached a westermost position, that is the proportion of the overprint component was minimum, and that of the ancient component maximum. The mean direction based on them is $D = 338°$, $I = 40°$, $k = 23$, $\alpha_{95} = 7°$, $N = 23$. This direction, though it clearly contains a sizable overprint component, may set a limit to the real angle of the rotation of Murge in the sense that the rotation can only be larger than the declination rotation measured at Spinazzola.

It is interesting to note that the direction we obtained for Spinazzola coincides with the one published by Channell & Tarling (1975) for Apulian bauxites.

At Pezza Rossa we also encountered an overprint problem. Due to the unremovable overprint in the red samples, we had to disregard them in the tectonic interpretation. Thus the locality mean direction (Table 1) is based on white samples only.

At Martine di Iatta the behaviour of the NRM on heating suggests a simple NRM. IRM acquisition curves, however, indicate a fairly high amount of goethite. Nevertheless, we have no indication for composite natural magnetization (above 225 °C) in any of the specimens studied from Martine di Iatta. Therefore, the locality is regarded as good for further tectonic interpretation, although the fold test is inconclusive. This is because the beds are nearly horizontal.

In Table 1, site and locality mean directions virtually free from secondary magnetization are compiled, both before and after tilt correction. Based on them, the Cenomanian–Turonian palaeomagnetic mean direction was calculated for Murge on a locality (before tectonic correction, $D = 327°$, $I = 44°$, $k = 55$, $\alpha_{95} = 17°$, $n = 3$; after tectonic correction, $D = 327°$, $I = 38°$, $k = 76$, $\alpha_{95} = 15°$) basis. The statistical parameters improve a little on correction. This is natural, since the dip angles of the strata are very low (Monte Scorzone: 6–7°, Martine di Iatta: 7–20°, Bivio: 14–25°, Pezza Rossa: 5–35°).

The disadvantage of not being able to make conclusive fold tests is compensated, however, by the fact that declination error is certainly not introduced by tilt correction (Tarling 1983) and tectonically undisturbed rocks are less prone to remagnetization than highly tectonized ones.

The area mean direction for Murge, based on three localities (12 sites, 56 samples) characterizing the Cenomanian–early Turonian are practically the same before and after tectonic correction. Nevertheless, to follow traditions in palaeomagnetism, tectonically corrected area mean direction will be interpreted in terms of tectonics.

CONCLUSIONS

As is clear from Table 1, we obtained statistically well-defined and reliable directions for 13 sites belonging to three localities. The sites at each locality represent different stratigraphic levels. Therefore, our results can be regarded as characteristic of the Cenomanian (including Turonian boundary) of Murge. In terms of tectonics they prove that the net rotation of Murge, since the Turonian, has been in the CCW sense.

Regarding the relationship of Murge to other areas of the Adriatic region, there is remarkable consistency between palaeomagnetic directions obtained for stable areas (Table 3). Determination of the exact position of Murge with respect to Africa is more difficult, since well-constrained palaeomagnetic poles for Africa are lacking.

One of the problems with African poles commonly reckoned with is the paucity of the direct data. However, palaeomagnetic poles for Africa may be estimated by combining poles for two or all Atlantic-bordering continents and reconstructing earlier plate positions (e.g. Van der Voo & French 1974; Westphal *et al.* 1986). Either calculated from direct African data or estimated with the help of reconstructions, the late Cretaceous palaeomagnetic poles published by different authors are fairly close (Lowrie 1986). Unfortunately, they are also very crude. In contrast to a stage pole, African poles represent a much longer time during which fast APW took place (Márton & Márton 1983). Thus the apparent difference between the palaeomagnetic pole for Murge and a late Cretaceous pole for

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Table 3. Summary of Cenomanian–Turonian palaeomagnetic data observed in stable areas of the Adriatic region.

	N	D°	I°	k	α95°	Pole lat	Pole lon	Reference
1.Istria	III	318	44	56	17	51	268	Márton E. & Veljovic 1983
2.Gargano	IV	328	38	65	12	56	258	VandenBerg 1983
	21	328	38	21	4			
3.Murge	III	327	38	55	17	56	261	present paper
	12	327	40	42	7			

N: Number of sampling sites (Arabic numbers) or sampling localities (Roman numerals). Statistical parameters (α_{95} and k) are based on the indicated N; D, I: mean palaeomagnetic declination and inclination, after tilt correction.

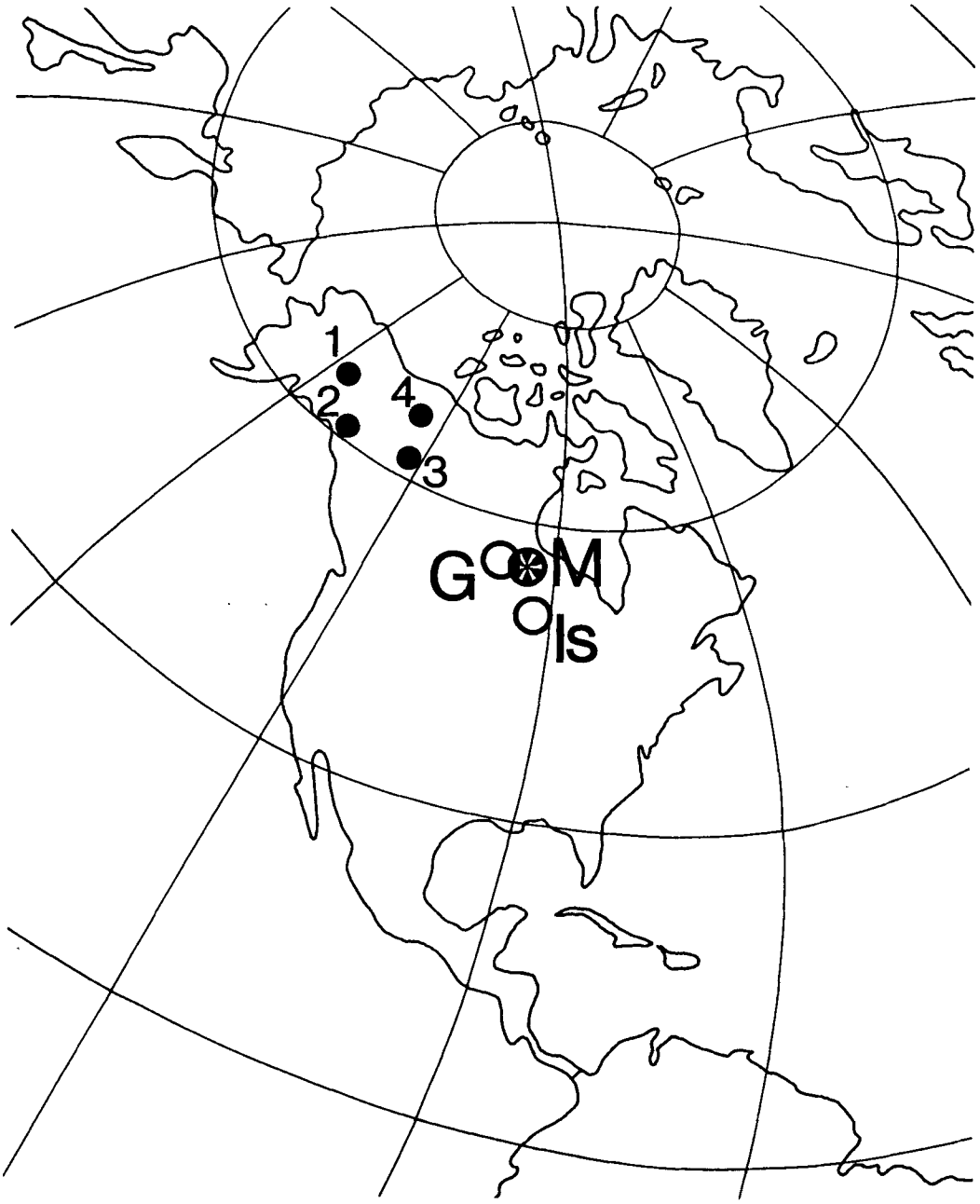


Figure 10. Cenomanian–Turonian palaeomagnetic pole position for Murge (M), Gargano (G) and Istria (Is), i.e. rigids areas of the Adriatic region compared with late Cretaceous poles for Africa. Poles for Africa are taken from a review by Lowrie (1986): 1 and 2 calculated from direct determinations, 3 and 4 computed with the help of reconstructions.

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Africa (Fig. 10) may be simply the consequence of a mismatch in time.

Nevertheless, Murge joins other stable areas of the Adriatic region (Istria and Gargano) via Cenomanian–Turonian poles, and Istria and Garagano have been shown to differ from Africa in the magnitude of CCW rotation for more than one geological stage (Márton & Veljovic 1983; VandenBerg 1983). Our new data from Murge, therefore, imply that an African Promontory in unchanged relative position with respect to Africa must be found further to the south. Unfortunately, the same data do not constrain precisely the timing of the rotation with respect to Africa.

There are palaeomagnetic studies claiming that both Gargano (Tozzi, Funicello & Parotto 1989) and Salentina Peninsula (Tozzi *et al.* 1988) were subjected to moderate CW rotation in post-Eocene time, but their rotation angle was different. However, this conclusion was based on palaeomagnetic directions that are close to the present field direction before tilt correction. The authors were aware of this problem, but argued that the reversal test and magnetite as carrier mineral supported the primary nature of the magnetization.

In the light of our observations for Bivio, namely that pure magnetite as carrier mineral may be secondary in Apulia, and the fact that the Eocene strata in Gargano as well as in Salento are in monoclinal position (i.e. fold test is impossible), we regard the Cenozoic history of the Apulian plate as an open question. Nevertheless, the evidence for net CCW rotation of the 'hard core' of the Adriatic region, with respect to Africa, since the late Cretaceous is established. Stable Istria–Gargano–Murge, after having separated from Africa, moved in a highly integrated manner, despite the lines of weakness between them.

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